

Description	Owners/Teams	Impact	Effort (FTE)	Milestones (M) and Decision Points (D) (year: description)	Risks and External Dependencies	Prob. of Applicability /Success/Adoption (Low, Medium, High)
Reconstruction						
2.34 Reconstruction: continuous code optimization	Development team, CMS ownership: reconstruction (J. Pata, C. Caputo)	~10% reco per-event time reduction per year until Run 4	0.5	'22-'28 M: code optimizations including compiler tuning and other core libraries	Dependencies: Profiler tools (e.g., IgProf) Risks: IgProf support continuity (or tool replacement). Algos and data structures may require major redesign for optimal use of hardware	Success: Reco L2s say High (L1s say Medium)
2.32 Reconstruction: Tracking displacement and pT cuts cost optimization [No updates from experts since LHCC Review]	Development team, CMS ownership: Tracking POG, reconstruction (J. Pata, C. Caputo)	~30% reco per-event time reduction	0.5 0.5	'22-'23 M: first set of optimized cuts '24-'25 D: final set of cuts before data taking, to be optimized with data	Risks: inability to cut time without degrading physics performance	Adoption: High
2.25 Reconstruction: mkFit tracking tool	Development team, CMS ownership: Tracking POG, reconstruction (J. Pata, C. Caputo), Marco Musich	~10% reco per-event time reduction (factor of 6 in track building)	2.0 2.0	'10-'23 M: commissioning '24-'25 D: Replace CKF track building in HL-LHC	Risks: CKF tracking implementation is a major change to a critical component of reconstruction	Success: High
2.17 Reconstruction: HGCal reconstruction with TICL	Development team, CMS ownership: HGCal DPG, reconstruction (J. Pata, C. Caputo)	HGCal < 5% of per-event reco time	2.0 2.0	'24 M: first implementation '25 D: production version		Success: High
3.8 Partial Prompt Reconstruction	CMS ownership: Computing Operations (C. Paus, A. Wightman), L1s, David Lange	~35/30% CPU savings @ 200/140 PU	0.2 0.1 0.1	'18 M: 10B b decays recorded '22 M: B parking at 3kHz '23-'25 M: exercise parking campaigns and associated tape recall exercises '32 D: "Emergency brake"	Risks: Delays in time to publication of physics papers	Adoption: hopefully Low (emergency brake in case other options fail)
Simulation						
2.9 Full G4-based application: incremental improvements to	Development team, CMS ownership: Simulation (Vladimir)	~50% G4-module per-event time	2	'22 D: list of further improvements during Run 3 '23 M: optimal Run 3 simulation '24 M: optimal Run 3 physics	Dependencies: '23 (M): Geant4 version 11.1 '24 (M): Geant4	Adoption of latest G4 versions: High Speedup success: High
2.11 Faster Simulation: incremental improvements of FastSim Monte Carlo chain, adopted by 50% of analyses	Development team, CMS ownership: Simulation (Vladimir Ivantchenko, Sam Bein)	6-10x faster sim + digi +recosim	3.0 3.0	'22-'23 M: development of physics improvements, mostly shower description '24-'25 M: implementation, integration, validation, optimization	Risks: adoption depends on improvements to physics, particularly, energy response, affecting MET	Success: Medium (50% chance of reaching the goal of 1/2 of analysis using FastSim)
Generation						
2.8.1 Code performance improvements of MC generators used by CMS [No updates from experts since LHCC Review]	Development team: community-wide physics generator authors, CMS GEN CMS ownership: generators (S. Bhattacharya, G. Chahal)	~10% gen per-event time reduction	–	External	Dependencies: external generator development groups	Success: High
2.8.2 Elimination of negative weight effect for the samples relevant to CMS [No updates from experts since LHCC Review]	Development team: community-wide physics generator authors, CMS GEN CMS ownership: generators (S. Bhattacharya, G. Chahal)	~30% of MC events needed for the same statistical power	–	External	Dependencies: external generator development groups	Success: High